

ACME ROBOTICS -- FIELD SERVICE ENGINEERING HANDBOOK (v3.2, internal)

1. Deployment Runbook

All production deployments to the fleet-control service go through a canary rollout: the new build first serves 5 percent of regional traffic for a minimum of twenty minutes before promotion. A canary is considered failed if error rate exceeds 2 percent or p99 latency exceeds 800 milliseconds during that window. If the canary fails, the on-call engineer must initiate a rollback within fifteen minutes of the failure being flagged; rollbacks beyond that window require a written incident postmortem regardless of customer impact. Deployments are only permitted Monday through Thursday before 16:00 local time, to keep a full business day of on-call coverage available in case of a slow-burn regression. Emergency hotfixes outside that window require sign-off from the on-call lead and the service owner.

2. Incident Response Policy

Incidents are classified SEV-1 through SEV-4 by customer impact. A SEV-1 incident -- full outage of the fleet-control API or any safety-critical robot command path -- requires the on-call engineer to acknowledge the page within five minutes and open a dedicated incident channel within ten minutes of acknowledgement. A SEV-2 incident, defined as degraded service for more than ten percent of fleets, carries a thirty-minute acknowledgement window. Every SEV-1 and SEV-2 incident requires a written postmortem within three business days of resolution, filed against the incident's tracking ticket and reviewed by the engineering lead before it is closed.

3. Database Migration Policy

Schema migrations against the fleet-telemetry database run through Flyway and must be backward compatible with the currently deployed application version for at least one full deploy cycle -- additive changes only (new nullable columns, new tables, new indexes); no column drops or type changes in the same migration that removes application code depending on the old shape. Every migration requires review and approval from two engineers, at least one of whom is not the migration's author, before it can run against the staging database. Migrations that touch a table larger than ten million rows must run during the Tuesday or Wednesday maintenance window and must include a tested rollback script in the same pull request.

4. On-Call Rotation & Escalation

The primary on-call rotation is one week long, handing off every Monday at 09:00 local time. Each engineer on the fleet-control team rotates through primary on-call roughly once every six weeks, assuming a team of six. If the primary on-call engineer does not acknowledge a page within ten minutes, the incident automatically escalates to the secondary on-call engineer; if the secondary does not acknowledge within a further ten minutes, it escalates to the engineering lead directly. Planned on-call swaps must be requested at least 48 hours in advance through the on-call calendar tool.

5. API Rate Limits

The public fleet-status API enforces a rate limit of 100 requests per minute per API key, with a burst allowance of 20 additional requests over any five-second window before throttling kicks in. Requests beyond the limit receive an HTTP 429 response with a Retry-After header. Enterprise-tier API keys may request a raised limit of up to 500 requests per minute by filing a capacity request with the platform team at least five business days before the increased limit is needed. Rate limit counters reset on a rolling sixty-second window per key, not a fixed calendar minute, so bursts near a minute boundary are still capped correctly.

6. Logging & Data Retention

Application logs from the fleet-control service are retained for 30 days in hot storage and then moved to cold storage for an additional 11 months before deletion, for a total retention period of 12 months. Logs containing customer location data must be redacted of precise GPS coordinates (rounded to the nearest 1 kilometre grid cell) before they leave hot storage. Access to unredacted location logs requires a documented business justification and sign-off from the data protection lead, logged in the access-request system.